TEMPERATURE

14.1 Thermal equilibrium

Candidates should be able to:

1 understand that (thermal) energy is transferred from a region of higher temperature to a region of lower temperature

2 understand that regions of equal temperature are in thermal equilibrium

14.2 Temperature scales

Candidates should be able to:

1 understand that a physical property that varies with temperature may be used for the measurement of temperature and state examples of such properties, including the density of a liquid, volume of a gas at constant pressure, resistance of a metal, e.m.f. of a thermocouple

2 understand that the scale of thermodynamic temperature does not depend on the property of any particular substance

3 convert temperatures between kelvin and degrees Celsius and recall that $T / \mathrm{K} = \theta / {}^\circ\mathrm{C} + 273.15$

4 understand that the lowest possible temperature is zero kelvin on the thermodynamic temperature scale and that this is known as absolute zero

14.3 Specific heat capacity and specific latent heat

Candidates should be able to:

1 define and use specific heat capacity

2 define and use specific latent heat and distinguish between specific latent heat of fusion and specific latent heat of vaporisation

1. M/J/07/Q2

(a) Use the kinetic theory of matter to explain why melting requires energy but there is no change in temperature.

....

....

....

.... [3]

(b) Define *specific latent heat of fusion*.

....

....

.... [2]

(c) A block of ice at 0 °C has a hollow in its top surface, as illustrated in Fig. 2.1.

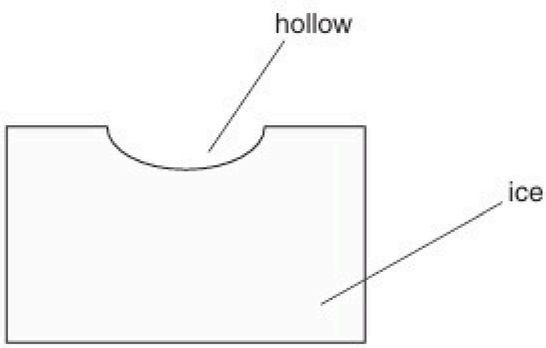


Fig. 2.1

A mass of 160 g of water at 100 °C is poured into the hollow. The water has specific heat capacity 4.20 kJ kg^{-1} K^{-1}. Some of the ice melts and the final mass of water in the hollow is 365 g.

(i) Assuming no heat gain from the atmosphere, calculate a value, in kJ kg^{-1}, for the specific latent heat of fusion of ice.

specific latent heat = ... kJ kg^{-1} [3]

(ii) In practice, heat is gained from the atmosphere during the experiment. This means that your answer to **(i)** is not the correct value for the specific latent heat.
State and explain whether your value in **(i)** is greater or smaller than the correct value.

....

....

.... [2]

2. O/N/08/Q2

(a) Define *specific latent heat of fusion.*

....

....

....[2]

(b) Some crushed ice at 0 °C is placed in a funnel together with an electric heater, as shown in Fig. 2.1.

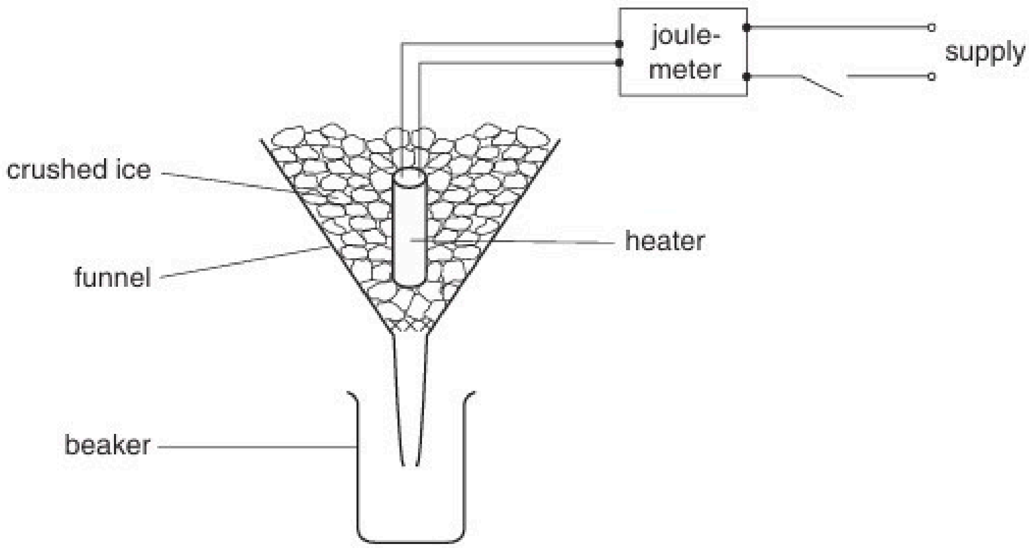


Fig. 2.1

The mass of water collected in the beaker in a measured interval of time is determined with the heater switched off. The mass is then found with the heater switched on. The energy supplied to the heater is also measured.
For both measurements of the mass, water is not collected until melting occurs at a constant rate.
The data shown in Fig. 2.2 are obtained.

	mass of water / g	energy supplied to heater / J	time interval / min
heater switched off	16.6	0	10.0
heater switched on	64.7	18000	5.0

Fig. 2.2

(i) State why the mass of water is determined with the heater switched off.

....

....[1]

(ii) Suggest how it can be determined that the ice is melting at a constant rate.

....

....[1]

(iii) Calculate a value for the specific latent heat of fusion of ice.

latent heat = ... kJ kg^{-1} [3]

3. M/J/09/Q3

When a liquid is boiling, thermal energy must be supplied in order to maintain a constant temperature.

(a) State two processes for which thermal energy is required during boiling.

1. ....

....

2. ....

.... [2]

(b) A student carries out an experiment to determine the specific latent heat of vaporisation of a liquid.
Some liquid in a beaker is heated electrically as shown in Fig. 3.1.

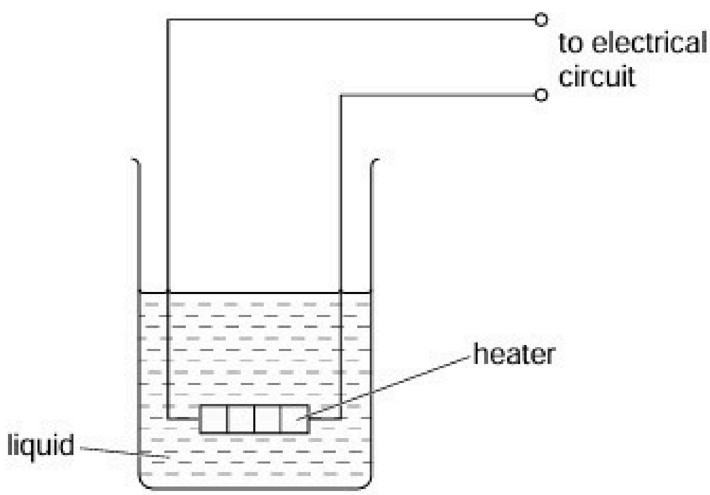


Fig. 3.1

Energy is supplied at a constant rate to the heater. When the liquid is boiling at a constant rate, the mass of liquid evaporated in 5.0 minutes is measured.
The power of the heater is then changed and the procedure is repeated.
Data for the two power ratings are given in Fig. 3.2.

power of heater /W	mass evaporated in 5.0 minutes /g
50.0	6.5
70.0	13.6

Fig. 3.2

(i) Suggest

1. how it may be checked that the liquid is boiling at a constant rate,

....

.... [1]

2. why the rate of evaporation is determined for two different power ratings.

....

.... [1]

(ii) Calculate the specific latent heat of vaporisation of the liquid.

specific latent heat of vaporisation = ... $J\,g^{-1}$ [3]

4. O/N/09/41/Q3

(a) A student states, quite wrongly, that temperature measures the amount of thermal energy (heat) in a body.

State and explain two observations that show why this statement is incorrect.

1. ....

....

....

2. ....

....

....

[4]

(b) A thermometer and an electrical heater are inserted into holes in an aluminium block of mass 960 g, as shown in Fig. 3.1.

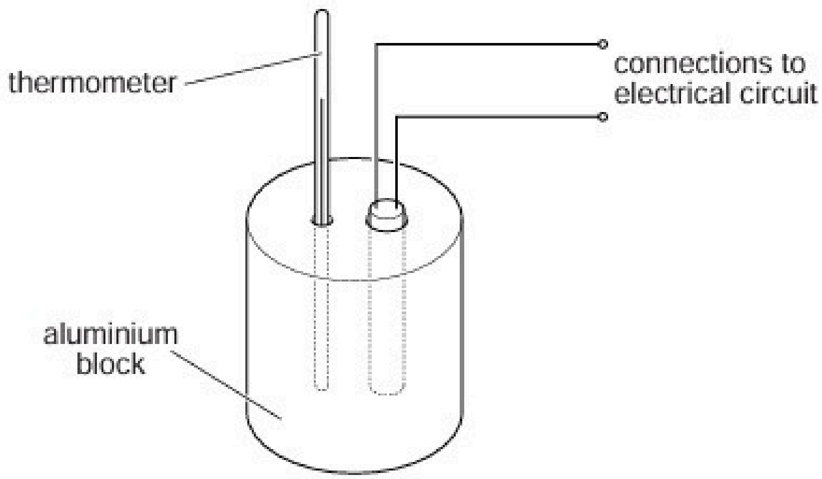


Fig. 3.1

The power rating of the heater is 54 W.

The heater is switched on and readings of the temperature of the block are taken at regular time intervals. When the block reaches a constant temperature, the heater is switched off and then further temperature readings are taken. The variation with time t of the temperature θ of the block is shown in Fig. 3.2.

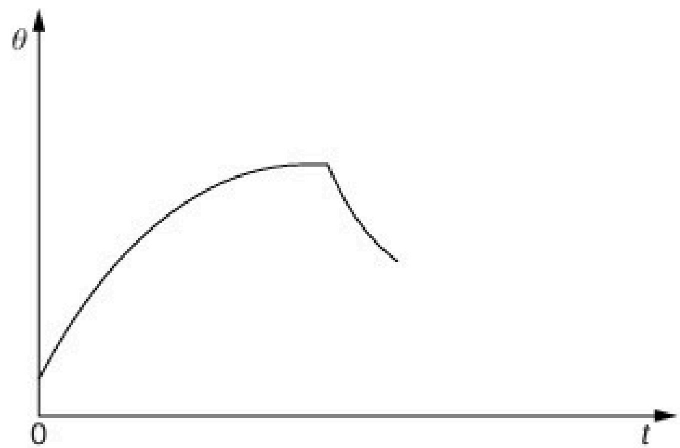


Fig. 3.2

(i) Suggest why the rate of rise of temperature of the block decreases to zero.

....

....

.... [2]

(ii) After the heater has been switched off, the maximum rate of fall of temperature is 3.7 K per minute.

Estimate the specific heat capacity of aluminium.

specific heat capacity = ... $\text{J kg}^{-1}\text{K}^{-1}$ [3]

5. M/J/10/42/Q3

(a) The resistance of a thermistor at 0 °C is 3840 Ω. At 100 °C the resistance is 190 Ω. When the thermistor is placed in water at a particular constant temperature, its resistance is 2300 Ω.

(i) Assuming that the resistance of the thermistor varies linearly with temperature, calculate the temperature of the water.

temperature = ... °C [2]

(ii) The temperature of the water, as measured on the thermodynamic scale of temperature, is 286 K.

By reference to what is meant by the thermodynamic scale of temperature, comment on your answer in **(i)**.

....

....

....

.... [3]

(b) A polystyrene cup contains a mass of 95 g of water at 28 °C.

A cube of ice of mass 12 g is put into the water. Initially, the ice is at 0 °C. The water, of specific heat capacity $4.2 \times 10^3\,\text{J kg}^{-1}\,\text{K}^{-1}$, is stirred until all the ice melts.

Assuming that the cup has negligible mass and that there is no heat exchange with the atmosphere, calculate the final temperature of the water.

The specific latent heat of fusion of ice is $3.3 \times 10^5\,\text{J kg}^{-1}$.

temperature = ... °C [4]

6. O/N/11/43/Q2

(a) A resistance thermometer and a thermocouple thermometer are both used at the same time to measure the temperature of a water bath.

Explain why, although both thermometers have been calibrated correctly and are at equilibrium, they may record different temperatures.

....

....

.... [2]

(b) State

(i) in what way the absolute scale of temperature differs from other temperature scales,

....

.... [1]

(ii) what is meant by the absolute zero of temperature.

....

.... [1]

(c) The temperature of a water bath increases from 50.00 °C to 80.00 °C.
Determine, in kelvin and to an appropriate number of significant figures,

(i) the temperature 50.00 °C,

temperature = ... K [1]

(ii) the change in temperature of the water bath.

temperature change = ... K [1]

_7. M/J/12/41/Q3_

(a) Define *specific latent heat.*

....

....

.... [2]

(b) The heater in an electric kettle has a power of 2.40 kW.
When the water in the kettle is boiling at a steady rate, the mass of water evaporated in 2.0 minutes is 106 g.
The specific latent heat of vaporisation of water is $2260\,\mathrm{J\,g^{-1}}$.

Calculate the rate of loss of thermal energy to the surroundings of the kettle during the boiling process.

rate of loss = ... W [3]

8. O/N/12/41/Q3

(a) Two metal spheres are in thermal equilibrium.
State and explain what is meant by *thermal equilibrium*.

....

....

.... [2]

(b) An electric water heater contains a tube through which water flows at a constant rate.
The water in the tube passes over a heating coil, as shown in Fig. 3.1.

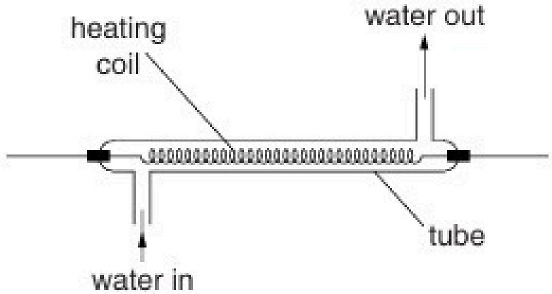


Fig. 3.1

The water flows into the tube at a temperature of 18 °C. When the power of the heater is 3.8 kW, the temperature of the water at the outlet is 42 °C.
The specific heat capacity of water is $4.2\,J\,g^{-1}\,K^{-1}$.

(i) Use the data to calculate the flow rate, in $g\,s^{-1}$, of water through the tube.

flow rate = ... $g\,s^{-1}$ [3]

(ii) State and explain whether your answer in **(i)** is likely to be an overestimate or an underestimate of the flow rate.

....

....

.... [2]

9. M/J/15/41/Q3

(a) Define *specific latent heat*.

....

....

....

.... [2]

(b) An electrical heater is immersed in some melting ice that is contained in a funnel, as shown in Fig. 3.1.

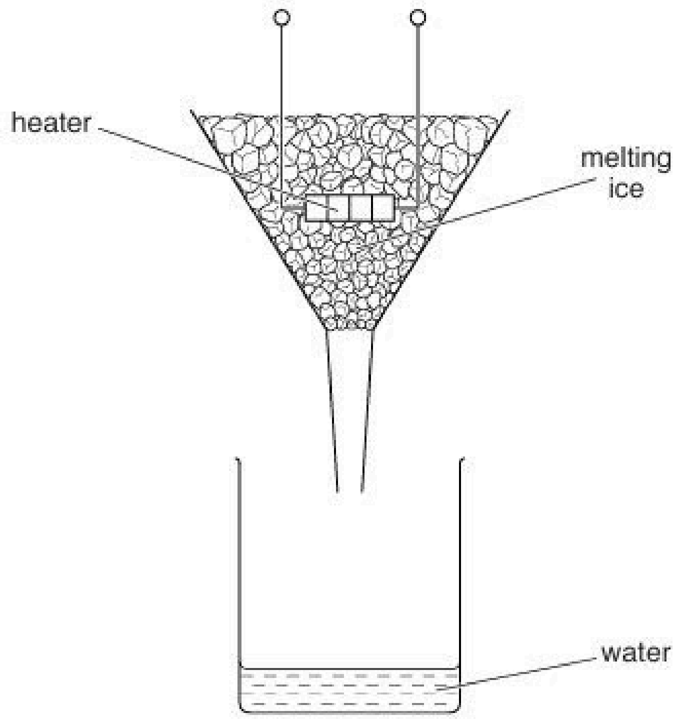


Fig. 3.1

The heater is switched on and, when the ice is melting at a constant rate, the mass m of ice melted in 5.0 minutes is noted, together with the power P of the heater. The power P of the heater is then increased. A new reading for the mass m of ice melted in 5.0 minutes is recorded when the ice is melting at a constant rate.

Data for the power P and the mass m are shown in Fig. 3.2.

power of heater P/W	mass m melted in 5.0 minutes/g	mass m melted per second/g s^{-1}
70	78	....
110	114	....

(i) Complete Fig. 3.2 to determine the mass melted per second for each power of the heater. [2]

(ii) Use the data in the completed Fig. 3.2 to determine

1. a value for the specific latent heat of fusion L of ice,

L = ... $J\,g^{-1}$ [3]

2. the rate h of thermal energy gained by the ice from the surroundings.

h = ... W [2]

10. M/J/15/42/Q3

(a) Define *specific latent heat.*

....

....

....[2]

(b) A beaker containing a liquid is placed on a balance, as shown in Fig. 3.1.

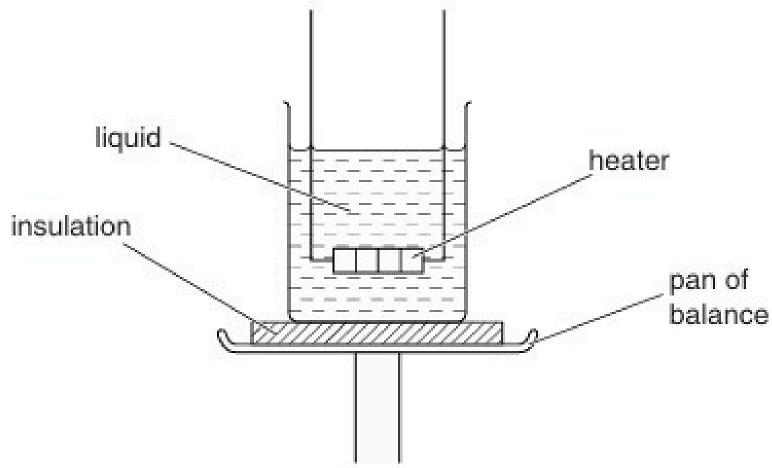
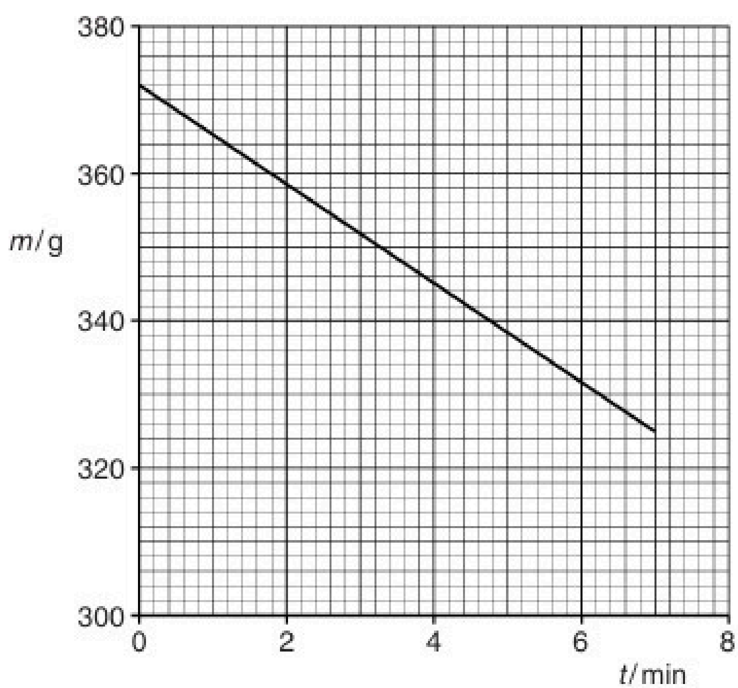


Fig. 3.1

A heater of power 110W is immersed in the liquid. The heater is switched on and, when the liquid is boiling, balance readings *m* are taken at corresponding times *t*.

A graph of the variation with time *t* of the balance reading *m* is shown in Fig. 3.2.



(i) State the feature of Fig. 3.2 which suggests that the liquid is boiling at a steady rate.

....

....[1]

(ii) Use data from Fig. 3.2 to determine a value for the specific latent heat *L* of vaporisation of the liquid.

L = ... $J\,kg^{-1}$ [3]

(iii) State, with a reason, whether the value determined in **(ii)** is likely to be an overestimate or an underestimate of the normally accepted value for the specific latent heat of vaporisation of the liquid.

....

....

....[2]

11. F/M/16/42/Q3

(a) Define *specific heat capacity*.

....

....

....[2]

(b) A student carries out an experiment to determine the specific heat capacity of a liquid using the apparatus illustrated in Fig. 3.1.

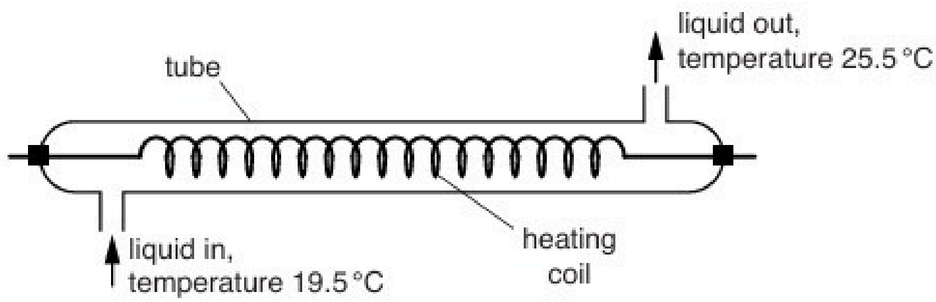


Fig. 3.1

Liquid enters the tube at a constant temperature of 19.5 °C and leaves the tube at a temperature of 25.5 °C. The mass of liquid flowing through the tube per unit time is *m*. Electrical power *P* is dissipated in the heating coil.

The student changes *m* and adjusts *P* until the final temperature of the liquid leaving the tube is 25.5 °C.

The data shown in Fig. 3.2 are obtained.

m / g s^{-1}	P/W
1.11	33.3
1.58	44.9

Fig. 3.2

(i) Suggest why the student obtains data for two values of *m*, rather than for one value.

....

....[1]

(ii) Calculate the specific heat capacity of the liquid.

Show your working.

specific heat capacity = ... $J\,kg^{-1}\,K^{-1}$ [3]

(c) When the heating coil in **(b)** dissipates 33.3 W of power, the potential difference V across the coil is given by the expression

$$V = 27.0\sin(395t).$$

The potential difference is measured in volts and the time t is measured in seconds.

Determine the resistance of the coil.

resistance = ... Ω [3]

[Total: 9]

12. M/J/16/42/Q3

(a) Explain what is meant by the statement that two bodies are in *thermal equilibrium*.

....

....

.... [1]

(b) Suggest suitable types of thermometer, one in each case, to measure

(i) the temperature of the flame of a Bunsen burner,

.... [1]

(ii) the change in temperature of a small crystal when it is exposed to a pulse of ultrasound energy.

.... [1]

(c) Some water is heated so that its temperature changes from 26.5 °C to a final temperature of 38.0 °C.

State, to an appropriate number of decimal places,

(i) the change in temperature in kelvin,

change = ... K [1]

(ii) the final temperature in kelvin.

final temperature = ... K [1]

[Total: 5]

13. O/N/17/41/Q1

(a) State

(i) what may be deduced from the difference in the temperatures of two objects,

....

.... [1]

(ii) the basic principle by which temperature is measured.

....

.... [1]

(b) By reference to your answer in **(a)(ii)**, explain why two thermometers may not give the same temperature reading for an object.

....

....

.... [2]

(c) A block of aluminium of mass 670 g is heated at a constant rate of 95 W for 6.0 minutes.
The specific heat capacity of aluminium is $910\,\mathrm{J\,kg^{-1}\,K^{-1}}$.
The initial temperature of the block is 24 °C.

(i) Assuming that no thermal energy is lost to the surroundings, show that the final temperature of the block is 80 °C.

[3]

(ii) In practice, there are energy losses to the surroundings.
The actual variation with time t of the temperature θ of the block is shown in Fig. 1.1.

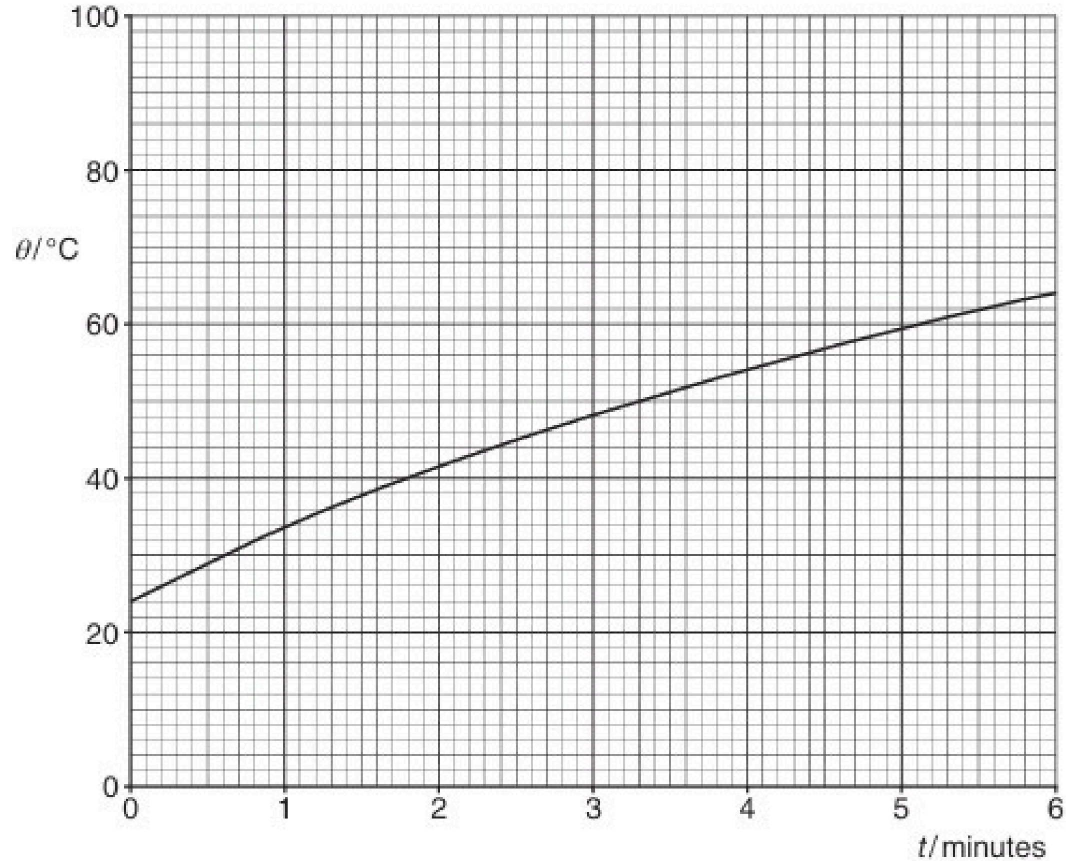


Fig. 1.1

1. Use the information in **(i)** to draw, on Fig. 1.1, a line to represent the temperature of the block, assuming no energy losses to the surroundings. [1]

2. Using Fig. 1.1, calculate the total energy loss to the surroundings during the heating process.

energy loss = ... J [2]

[Total: 10]

14. O/N/17/42/Q2

(a) State what is meant by *specific latent heat*.

....

....

....[2]

(b) A beaker of boiling water is placed on the pan of a balance, as illustrated in Fig. 2.1.

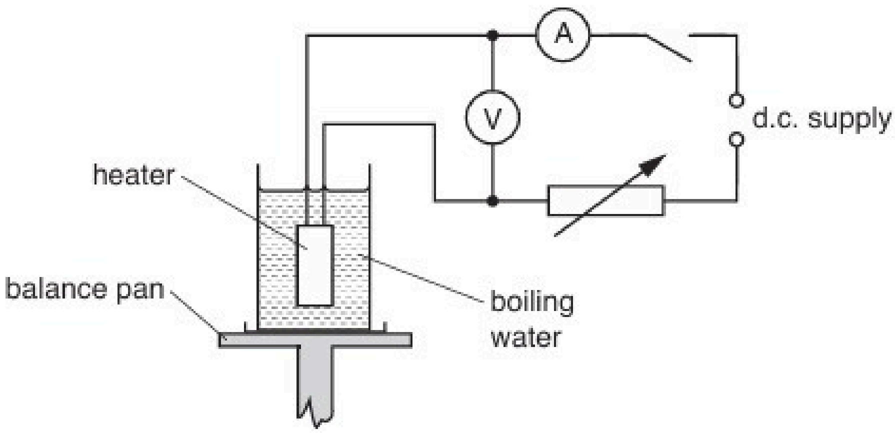


Fig. 2.1

The water is maintained at its boiling point by means of a heater.
The change M in the balance reading in 300 s is determined for two different input powers to the heater.
The results are shown in Fig. 2.2.

voltmeter reading / V	ammeter reading / A	M / g
11.5	5.2	5.0
14.2	6.4	9.1

Fig. 2.2

(i) Energy is supplied continuously by the heater.
State where, in this experiment,

1. external work is done,

....

....

2. internal energy increases. Explain your answer.

....

....

....
[3]

(ii) Use data in Fig. 2.2 to determine the specific latent heat of vaporisation of water.

specific latent heat = ... $J g^{-1}$ [3]

[Total: 8]

15. M/J/18/42/Q3

(a) During melting, a solid becomes liquid with little or no change in volume.

Use kinetic theory to explain why, during the melting process, thermal energy is required although there is no change in temperature.

....

....

....

....

....[3]

(b) An aluminium can of mass 160 g contains a mass of 330 g of warm water at a temperature of 38 °C, as illustrated in Fig. 3.1.

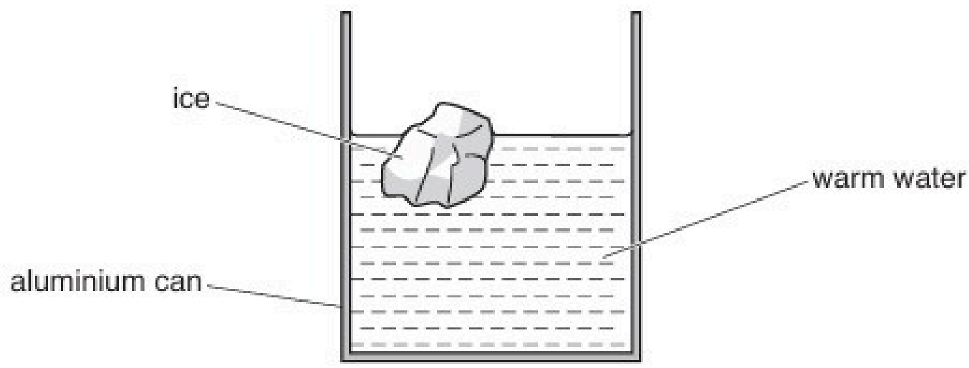


Fig. 3.1

A mass of 48 g of ice at –18 °C is taken from a freezer and put in to the water. The ice melts and the final temperature of the can and its contents is 23 °C.

Data for the specific heat capacity *c* of aluminium, ice and water are given in Fig. 3.2.

	c / J g^{-1} K^{-1}
aluminium	0.910
ice	2.10
water	4.18

Fig. 3.2

Assuming no exchange of thermal energy with the surroundings,

(i) show that the loss in thermal energy of the can and the warm water is 2.3×10^4 J,

[2]

(ii) use the information in **(i)** to calculate a value L for the specific latent heat of fusion of ice.

L = ... $\mathrm{J\,g^{-1}}$ [2]

[Total: 7]

16. O/N/18/42/Q3

(a) Define *specific latent heat of fusion*.

....

....

....[2]

(b) A student sets up the apparatus shown in Fig. 3.1 in order to investigate the melting of ice.

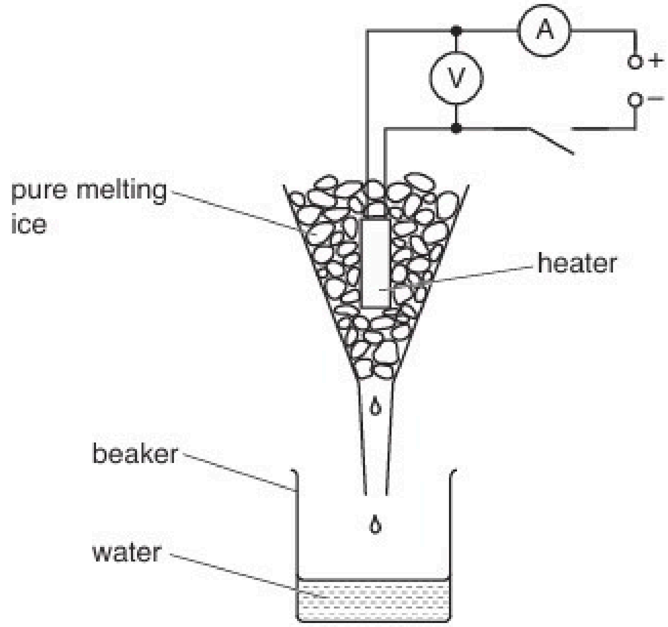


Fig. 3.1

The heater is switched on.
When the pure ice is melting at a constant rate, the data shown in Fig. 3.2 are collected.

voltmeter reading /V	ammeter reading /A	initial mass of beaker plus water/g	final mass of beaker plus water/g	time of collection /minutes
12.8	4.60	121.5	185.0	5.00

Fig. 3.2

The specific latent heat of fusion of ice is $332\,J\,g^{-1}$.

(i) State what is observed by the student that shows that the ice is melting at a constant rate.

....

....[1]

(ii) Use the data in Fig. 3.2 to determine the rate at which

1. thermal energy is transferred to the melting ice,

rate = ... W

2. thermal energy is gained from the surroundings.

rate = ... W
[4]

[Total: 7]

17. O/N/19/41/Q3

(a) State what is meant by *specific latent heat*.

....

....

.... [2]

(b) A student determines the specific latent heat of vaporisation of a liquid using the apparatus illustrated in Fig. 3.1.

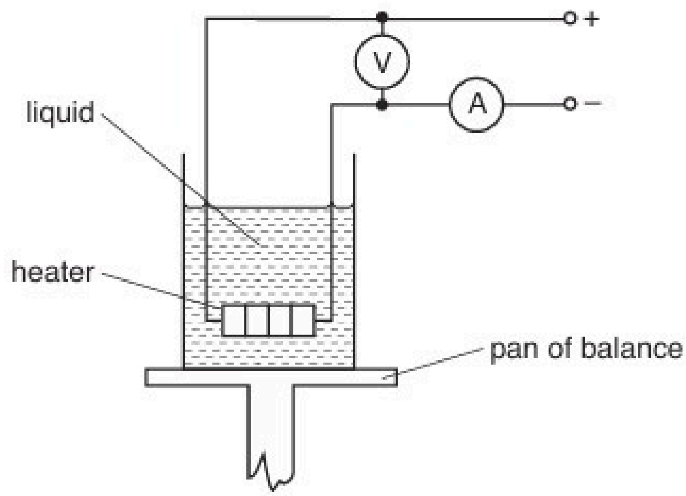


Fig. 3.1

The heater is switched on. When the liquid is boiling at a constant rate, the balance reading is noted at 2.0 minute intervals.

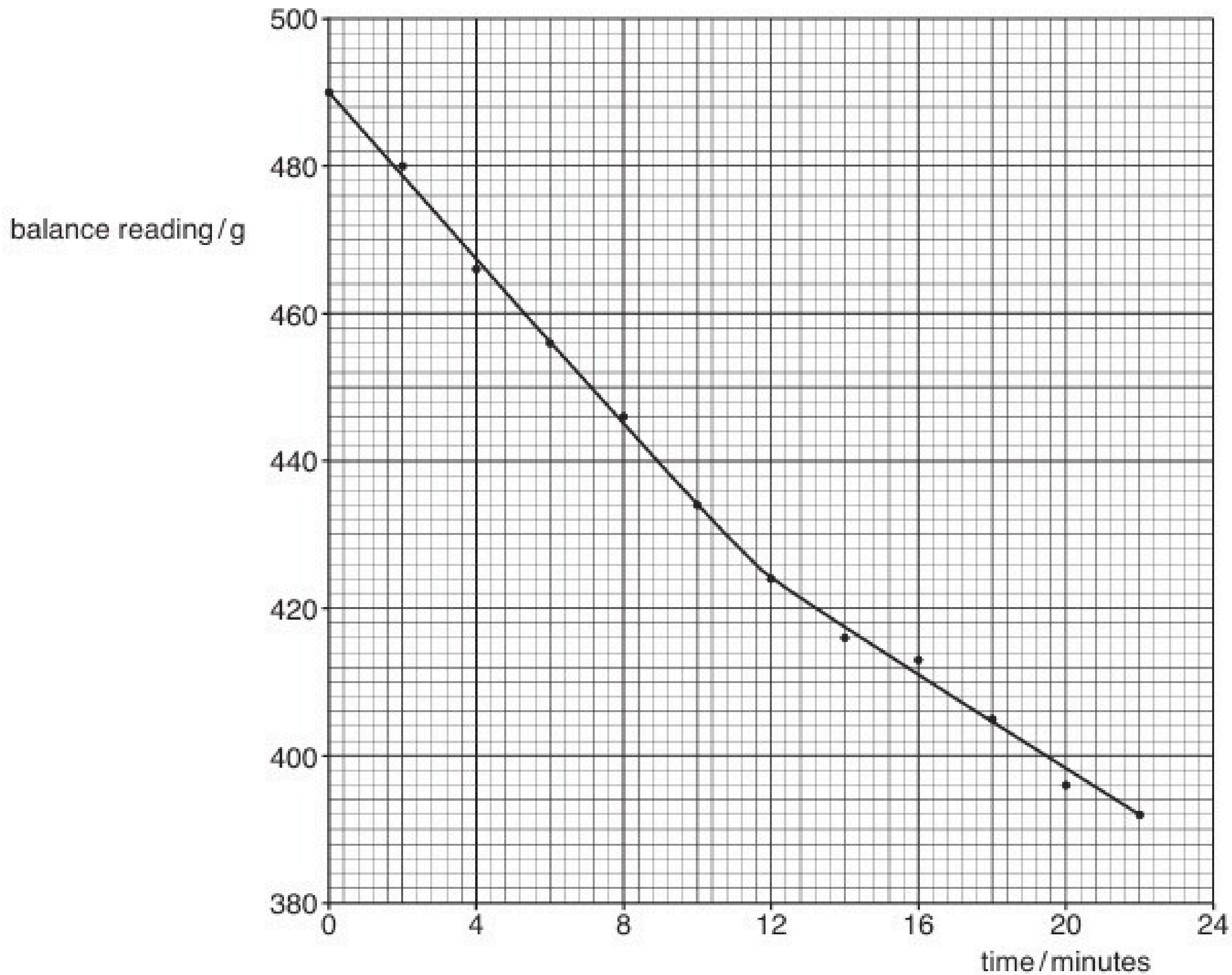
After 10 minutes, the current in the heater is reduced and the balance readings are taken for a further 12 minutes.

The readings of the ammeter and of the voltmeter are given in Fig. 3.2.

	ammeter reading /A	voltmeter reading /V
from time 0 to time 10 minutes	1.2	230
after time 10 minutes	1.0	190

Fig. 3.2

The variation with time of the balance reading is shown in Fig. 3.3.



(i) From time 0 to time 10.0 minutes, the mass of liquid evaporated is 56 g.

Use Fig. 3.3 to determine the mass of liquid evaporated from time 12.0 minutes to time 22.0 minutes.

mass = ... g [1]

(ii) Explain why, although the power of the heater is changed, the rate of loss of thermal energy to the surroundings may be assumed to be constant.

....

.... [1]

(iii) Determine a value for the specific latent heat of vaporisation L of the liquid.

L = ... $\mathrm{J\,g^{-1}}$ [4]

(iv) Calculate the rate at which thermal energy is transferred to the surroundings.

rate = ... W [2]

[Total: 10]

18. O/N/19/42/Q3

(a) State what is meant by *specific latent heat*.

....

....

.... [2]

(b) A student uses the apparatus illustrated in Fig. 3.1 to determine a value for the specific latent heat of fusion of ice.

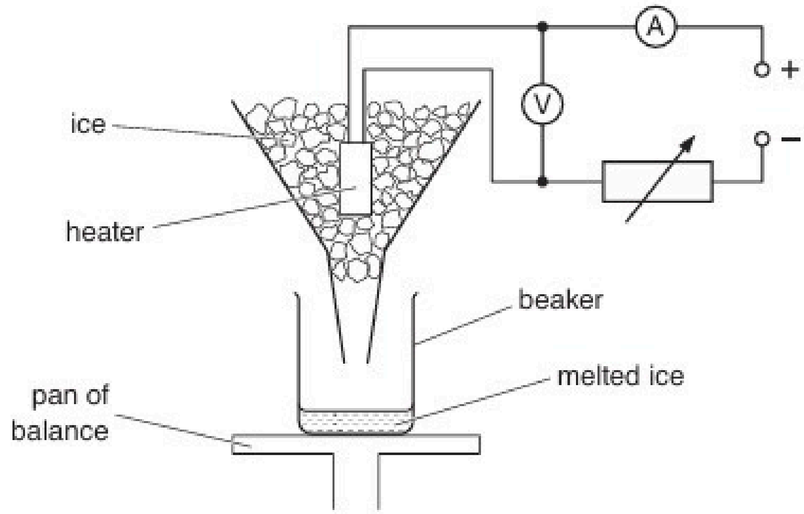


Fig. 3.1

The balance reading measures the mass of the beaker and the melted ice (water) in the beaker.

The heater is switched on and pieces of ice at 0 °C are added continuously to the funnel so that the heater is always surrounded by ice.

When water drips out of the funnel at a constant rate, the balance reading is noted at 2.0 minute intervals. After 10 minutes, the current in the heater is increased and the balance readings are taken for a further 12 minutes.

The variation with time of the balance reading is shown in Fig. 3.2.

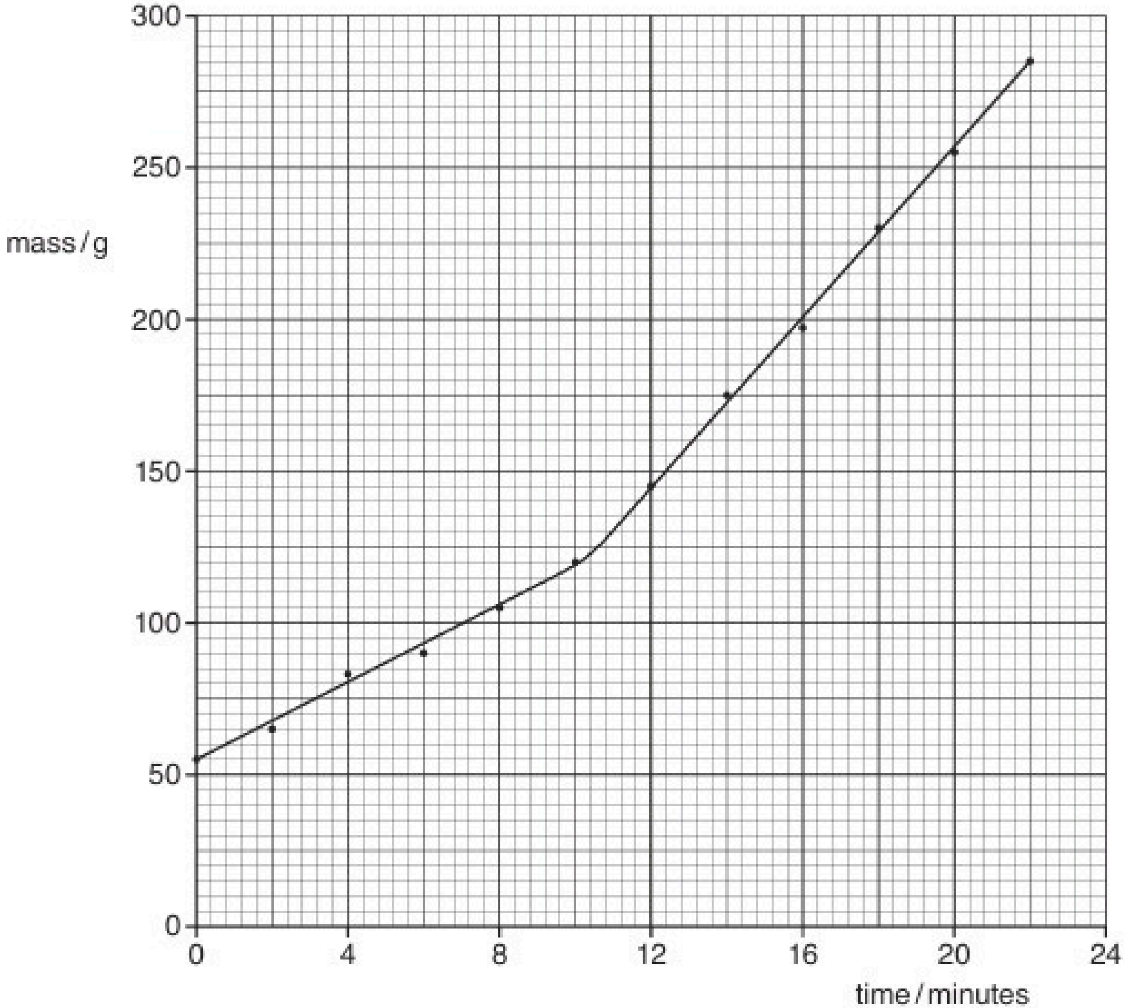


Fig. 3.2

The readings of the ammeter and of the voltmeter are shown in Fig. 3.3.

	ammeter reading / A	voltmeter reading / V
from time 0 to time 10 minutes	1.8	7.3
after time 10 minutes	3.6	15.1

Fig. 3.3

(i) From time 0 to time 10.0 minutes, 65 g of ice is melted.

Use Fig. 3.2 to determine the mass of ice melted from time 12.0 minutes to time 22.0 minutes.

mass = ... g [1]

(ii) Explain why, although the power of the heater is changed, the rate at which thermal energy is transferred from the surroundings to the ice is constant.

....

.... [1]

(iii) Determine a value for the specific latent heat of fusion L of ice.

L = ... $J g^{-1}$ [4]

(iv) Calculate the rate at which thermal energy is transferred from the surroundings to the ice.

rate = ... W [2]

[Total: 10]

19. F/M/21/42/Q3

(a) Using a simple kinetic model of matter, describe the structure of a solid.

....

....

.... [2]

(b) The specific latent heat of vaporisation is much greater than the specific latent heat of fusion for the same substance.
Explain this, in terms of the spacing of molecules.

....

....

.... [1]

(c) A heater supplies energy at a constant rate to 0.045 kg of a substance. The variation with time of the temperature of the substance is shown in Fig. 3.1. The substance is perfectly insulated from its surroundings.

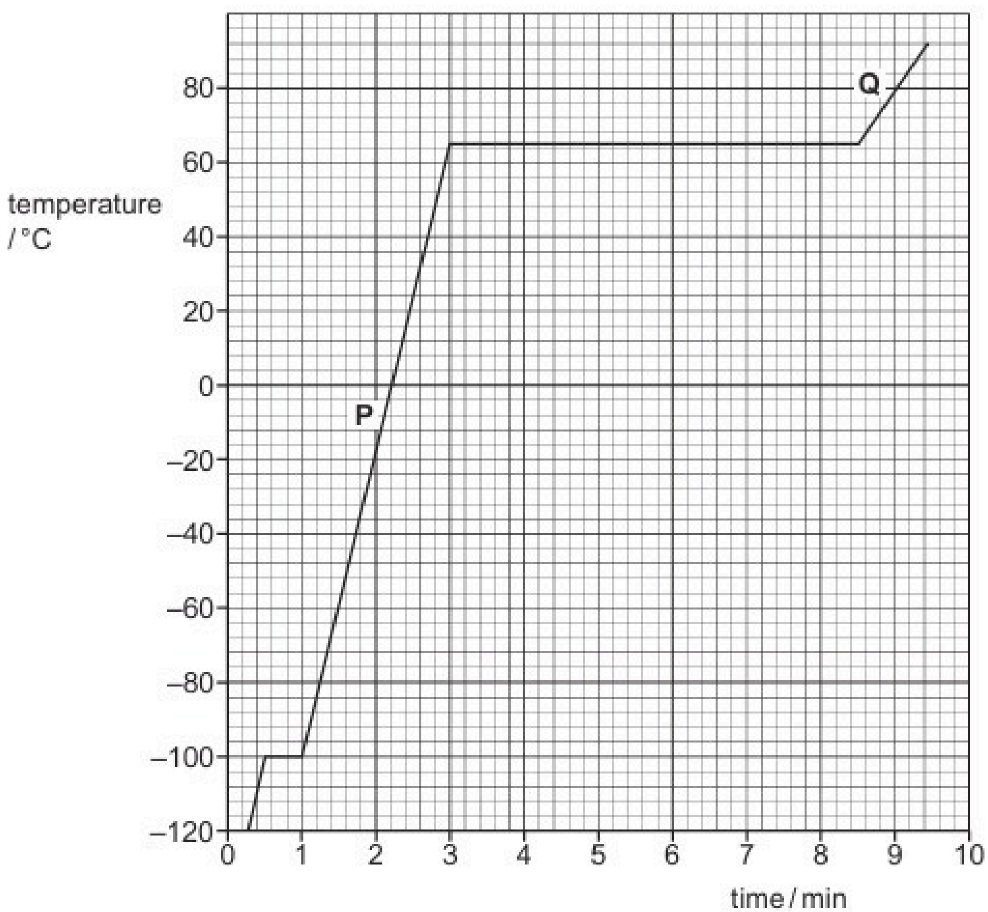


Fig. 3.1

(i) Determine the temperature at which the substance melts.

temperature = ... °C [1]

(ii) The power of the heater is 150 W.
Use data from Fig. 3.1 to calculate, in $kJ\,kg^{-1}$, the specific latent heat of vaporisation L of the substance.

L = ... $kJ\,kg^{-1}$ [3]

(iii) Suggest what can be deduced from the fact that section **Q** on the graph is less steep than section **P**.

....

.... [1]

[Total: 8]

20. O/N/22/41/Q2

Fig. 2.1 shows a laboratory thermometer that is calibrated to measure temperature in degrees Celsius.

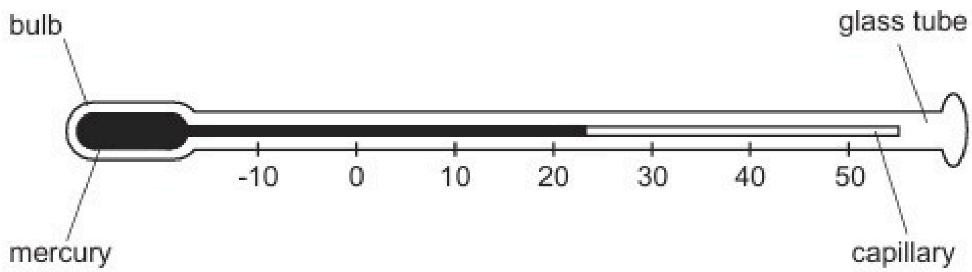


Fig. 2.1

The thermometer makes use of the fact that the density of mercury varies with temperature.

(a) State **two** other physical properties of materials, apart from the density of a liquid, that can be used for measuring temperature.

1 ....

2 ....
[2]

(b) The thermometer is initially at 23.0 °C, as shown in Fig. 2.1. It is used to measure the temperature of an insulated beaker of water that is at 37.4 °C. The bulb of the thermometer is inserted into the water, and the water is stirred until the reading on the thermometer becomes steady.

The mass of water in the beaker is 18.7 g.
The mass of mercury in the thermometer is 6.94 g.
The specific heat capacity of water is $4.18\,J\,g^{-1}\,K^{-1}$.
The specific heat capacity of mercury is $0.140\,J\,g^{-1}\,K^{-1}$.
The glass of the thermometer and the beaker containing the water can be considered to have negligible heat capacity.

(i) Calculate, to three significant figures, the final steady temperature indicated by the thermometer in the water.

temperature = ... °C [4]

(ii) Suggest **one** change that could be made to the design of the thermometer that would enable it to give a more accurate measurement of temperature.

....

.... [1]

(c) **(i)** Explain why the thermometer in Fig. 2.1 does **not** provide a direct measurement of thermodynamic temperature.

....

....

.... [2]

(ii) Thermodynamic temperature T may be determined by the behaviour of a type of substance for which T is proportional to the product of pressure and volume.

State the name of this type of substance.

.... [1]

[Total: 10]

21. M/J/23/41/Q3

(a) State the reason why two objects that are at the same temperature are described as being in thermal equilibrium.

....

.... [1]

(b) Fig. 3.1 shows the variations with temperature of the densities of mercury and of water between 0 °C and 100 °C.

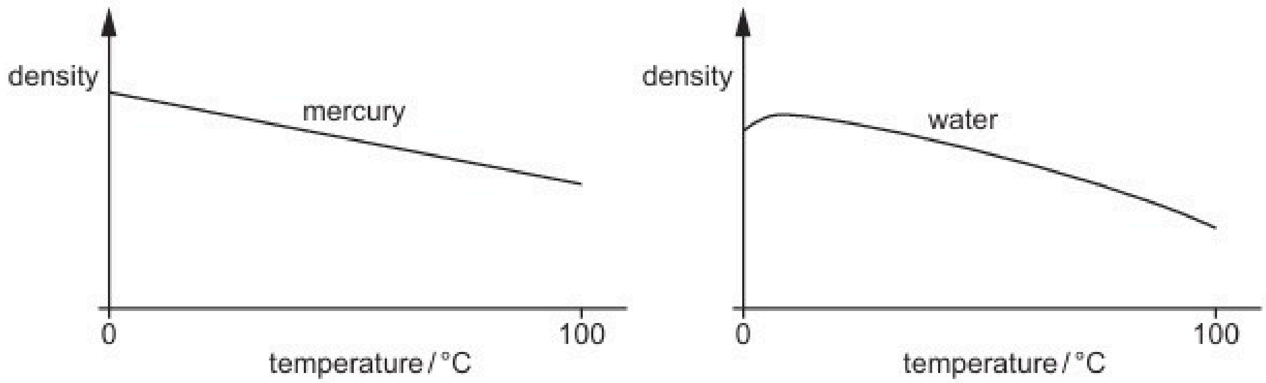


Fig. 3.1

Temperature may be measured using the variation with temperature of the density of a liquid.

Suggest why, for measuring temperature over this temperature range:

(i) mercury is a suitable liquid

....

.... [1]

(ii) water is not a suitable liquid.

....

....

.... [2]

(c) A beaker contains a liquid of mass 120 g. The liquid is supplied with thermal energy at a rate of 810 W. The beaker has a mass of 42 g and a specific heat capacity of 0.84 $J g^{-1} K^{-1}$. The beaker and the liquid are in thermal equilibrium with each other at all times and are insulated from the surroundings.

Fig. 3.2 shows the variation with time *t* of the temperature of the liquid.

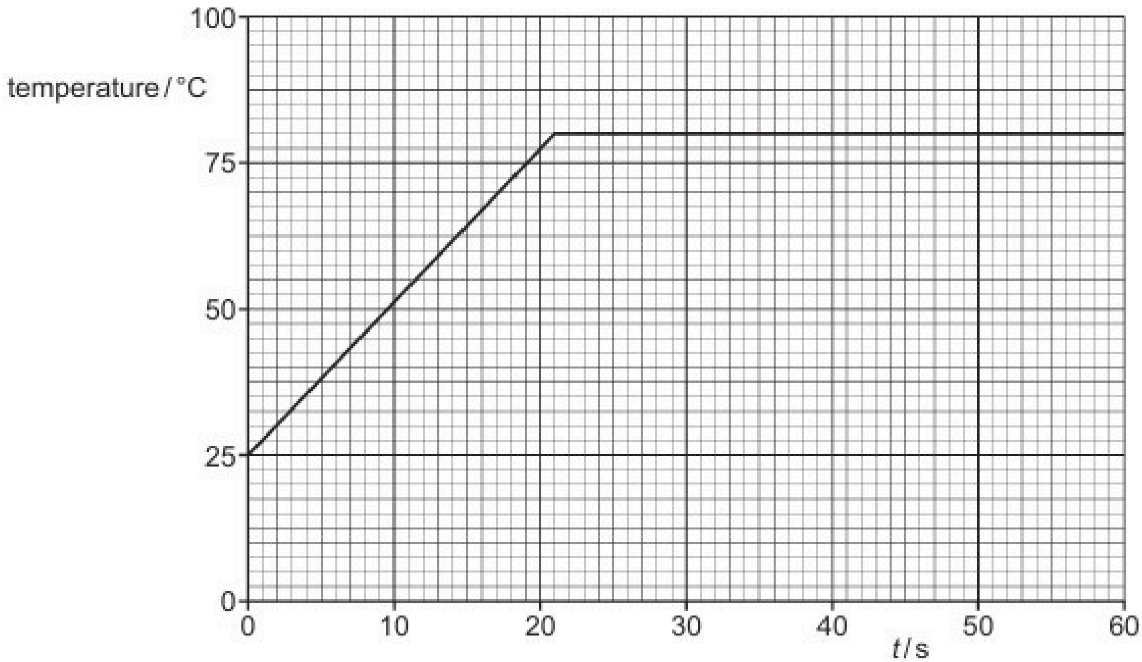


Fig. 3.2

(i) State the boiling temperature, in °C, of the liquid.

temperature = ... °C [1]

(ii) Determine the specific heat capacity, in $J\,g^{-1}\,K^{-1}$, of the liquid.

specific heat capacity = ... $J\,g^{-1}\,K^{-1}$ [4]

(d) The experiment in **(c)** is repeated using water instead of the liquid in **(c)**. The mass of liquid used, the power supplied, and the initial temperature are all unchanged.
The specific heat capacity of water is approximately twice that of the liquid in **(c)**.
The boiling temperature of water is 100 °C.

On Fig. 3.2, sketch the variation with time t of the temperature of the water between t = 0 and t = 60 s. Numerical calculations are not required. [2]